CHAPTER 7

EXCRETORY PRODUCTS AND THEIR ELIMINATION

CHAPTER ESSENCE AT A GLANCE:
Nephron filtration, counter-current medullary concentration, and neuro-hormonal feedback regulating osmoregulation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Nitrogenous Wastes and Excretory Patterns:** Animals excrete nitrogenous wastes according to water availability: Ammonotelic (ammonia, highly toxic, highly soluble, requires large amounts of water, e.g., bony fishes, aquatic amphibians); Ureotelic (urea, less toxic, synthesized in liver through ornithine cycle, e.g., mammals, terrestrial amphibians); Uricotelic (uric acid, least toxic, insoluble, excreted as paste or pellet with minimal water loss, e.g., birds, reptiles, insects).

• **Nephron Filtration and Tubular Exchange:** Urine formation comprises three steps: (1) Glomerular filtration: 1100 to 1200 mL of blood filtered per minute through glomerular capillaries across podocyte filtration slits (GFR = 125 mliters per minute = 180 liters per day); (2) Tubular reabsorption: 99% of filtrate reabsorbed in PCT (glucose, amino acids actively, water obligatorily); (3) Tubular secretion: H+, K+, and ammonia secreted into tubules to maintain blood pH and ionic balance.

• **Counter-Current Multiplier Mechanism:** Henle's loop and vasa recta flow in opposite directions to establish an osmotic medullary gradient: the descending limb of Henle's loop is permeable to water and impermeable to electrolytes (filtrate becomes hypertonic); the ascending limb is impermeable to water and actively transports NaCl into the medullary interstitium; interstitial osmolarity rises from 300 mOsmol per liter in cortex to 1200 mOsmol per liter in inner medulla.

• **Hormonal Regulation: ADH and RAAS:** Osmoreceptors trigger posterior pituitary to secrete ADH (Vasopressin), promoting water reabsorption in DCT and collecting duct. Juxtaglomerular Apparatus (JGA): low GFR triggers JG cells to secrete Renin, converting Angiotensinogen to Angiotensin I, then Angiotensin II (vasoconstrictor, stimulates adrenal cortex to release Aldosterone for Na+ and water reabsorption) = RAAS. High blood pressure stimulates heart atria to release ANF, which causes vasodilation and checks RAAS.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{GFR} = 125\ \text{mL min}^{-1} \rightarrow 180\ \text{L day}^{-1} \quad [99\%\ \text{reabsorbed} \rightarrow 1.5\ \text{L urine}]$$

Glomerular Filtration Rate and Daily Tubular Reabsorption

$$\text{Medullary Gradient} = 300\ \text{mOsmol L}^{-1}\ (\text{Cortex}) \longrightarrow 1200\ \text{mOsmol L}^{-1}\ (\text{Inner Medulla})$$

Osmotic Counter-Current Medullary Concentration Gradient

$$\text{RAAS Axis} = \text{Renin (JGA)} \rightarrow \text{Angiotensin II} \rightarrow \text{Aldosterone} \rightarrow \text{Na}^{+}\ \text{and Water Uptake}$$

Renin-Angiotensin-Aldosterone System Cascade

$$\text{Filtration Barrier} = \text{Capillary Endothelium} + \text{Basement Membrane} + \text{Podocyte Slit Pores}$$

Ultrafiltration Tripartite Microscopic Architecture

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The descending limb of Henle's loop is permeable to water and impermeable to electrolytes; the ascending limb is impermeable to water and permeable to electrolytes.

• **Alert:** ANF (Atrial Natriuretic Factor) acts antagonistically to the RAAS mechanism: ANF causes vasodilation and promotes sodium excretion, lowering blood pressure.

• **Alert:** Uremia is the accumulation of urea in blood due to kidney malfunction, which is medically treated by hemodialysis or kidney transplantation.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Compare ammonotelism, ureotelism, and uricotelism based on the toxicity of the excretory product and the volume of water required for excretion. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Ammonotelism: Ammonia is the most toxic nitrogenous waste and requires very large volumes of water for its excretion. **[0.75 Mark]**

• Ureotelism: Urea is moderately toxic, requires moderate amounts of water, and is excreted by mammals and amphibians. **[0.75 Mark]**

• Uricotelism: Uric acid is the least toxic, insoluble in water, and excreted as a thick pellet or paste with minimal water loss in birds and reptiles. **[0.5 Mark]**

2. What is uremia? Name the artificial clinical procedure utilized to remove nitrogenous waste products from the blood of uremic patients. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Uremia is a clinical condition characterized by the abnormal accumulation of urea and other nitrogenous wastes in the blood due to kidney failure. **[1 Mark]**

• The clinical procedure utilized is hemodialysis (using an artificial dialyzing unit with cellophane coiled tubes). **[1 Mark]**

3. Describe the three sequential physiological steps involved in urine formation within a human nephron. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Glomerular ultrafiltration: Blood is filtered through three layers of the filtration membrane into Bowman's capsule under hydrostatic pressure, producing about 180 liters of protein-free filtrate per day. **[1 Mark]**

• Tubular reabsorption: Nearly 99 percent of the filtrate is reabsorbed by active transport (glucose, amino acids, Na+) and passive osmosis (water, urea) across the renal tubules. **[1 Mark]**

• Tubular secretion: Tubular cells actively secrete substances like hydrogen ions, potassium ions, and ammonia into the filtrate to maintain ionic balance and blood pH. **[1 Mark]**

4. Explain the counter-current mechanism operating between Henle's loop and the vasa recta in concentrating urine in the renal medulla. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Counter-current arrangement: The flow of filtrate in the two limbs of Henle's loop and the flow of blood in the two limbs of vasa recta are in opposite directions. **[1 Mark]**

• Henle's loop action: Ascending limb actively pumps NaCl into the medullary interstitium, which is carried away by the descending vasa recta; water diffuses out of the descending limb of Henle's loop. **[1 Mark]**

• Gradient creation: This maintains a high osmolarity gradient from 300 mOsmol per liter in the renal cortex to 1200 mOsmol per liter in the inner medulla, enabling collecting ducts to reabsorb water and produce concentrated urine. **[1 Mark]**

5. Outline the Renin-Angiotensin-Aldosterone System (RAAS) mechanism initiated by juxtaglomerular cells in response to a drop in glomerular blood pressure. [Score: 3 Marks]

Answer & Key Valuation Steps:

• A fall in glomerular blood flow or GFR activates juxtaglomerular (JG) cells to release the enzyme renin into the bloodstream. **[1 Mark]**

• Renin converts plasma angiotensinogen into angiotensin I, and subsequently into angiotensin II, which is a powerful vasoconstrictor that elevates glomerular blood pressure. **[1 Mark]**

• Angiotensin II also stimulates the adrenal cortex to release aldosterone, which causes active reabsorption of Na+ and water from distal tubules, restoring blood volume and GFR. **[1 Mark]**

6. Explain how Antidiuretic Hormone (ADH) and Atrial Natriuretic Factor (ANF) act antagonistically in regulating blood volume and osmolality. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• ADH action: Released by the posterior pituitary during water loss or high blood osmolality; facilitates water reabsorption from distal tubules and collecting ducts, preventing diuresis and raising blood pressure. **[1.5 Marks]**

• ANF action: Released by heart atria when blood pressure rises; causes vasodilation of blood vessels and inhibits renin secretion, promoting sodium excretion to lower blood pressure, acting as an antagonist to ADH. **[1.5 Marks]**

7. A patient diagnosed with Diabetes Insipidus excretes large volumes of dilute urine (polyuria) and experiences extreme thirst (polydipsia), but has normal blood glucose levels. Identify the deficient hormone, name the endocrine gland responsible, and explain why collecting ducts fail to concentrate urine. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Deficient hormone: Antidiuretic Hormone (ADH) or Vasopressin. **[1 Mark]**

• Endocrine gland: Synthesized by the Hypothalamus and released into circulation by the Posterior Pituitary (Neurohypophysis). **[1 Mark]**

• Pathophysiological explanation: ADH normally binds to V2 receptors in the principal cells of distal tubules and collecting ducts, inserting aquaporin water channels into apical membranes to permit water reabsorption. **[1 Mark]**

• Clinical consequence: In the absence of ADH, the collecting duct walls remain completely impermeable to water, so hypotonic tubular fluid passes directly to the bladder as copious dilute urine. **[1 Mark]**